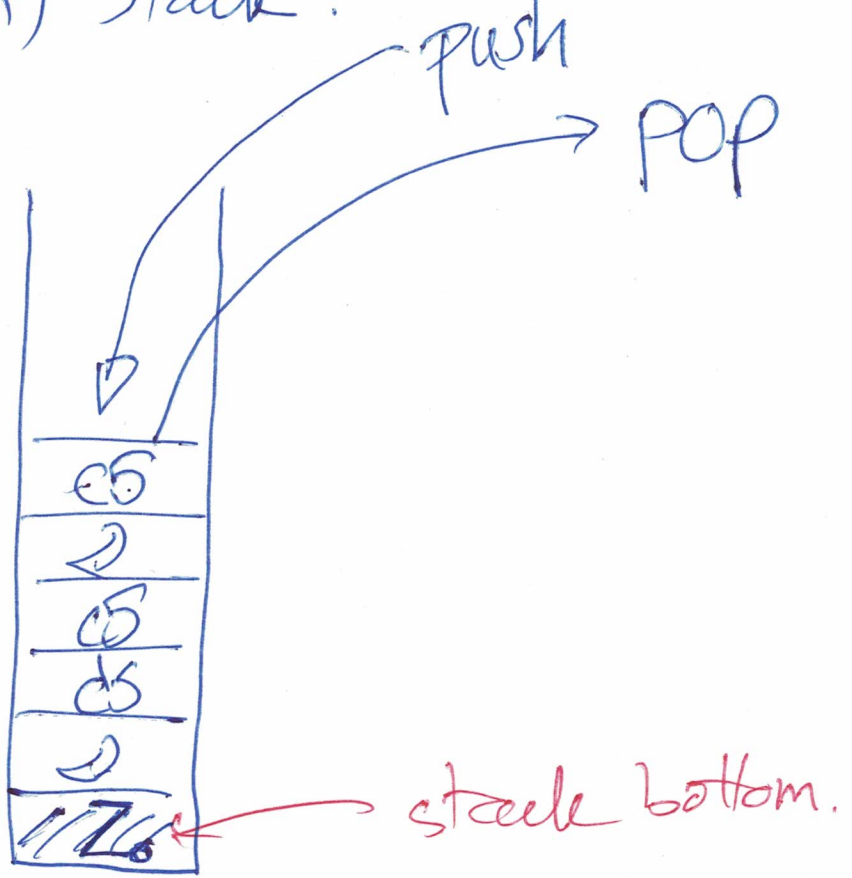
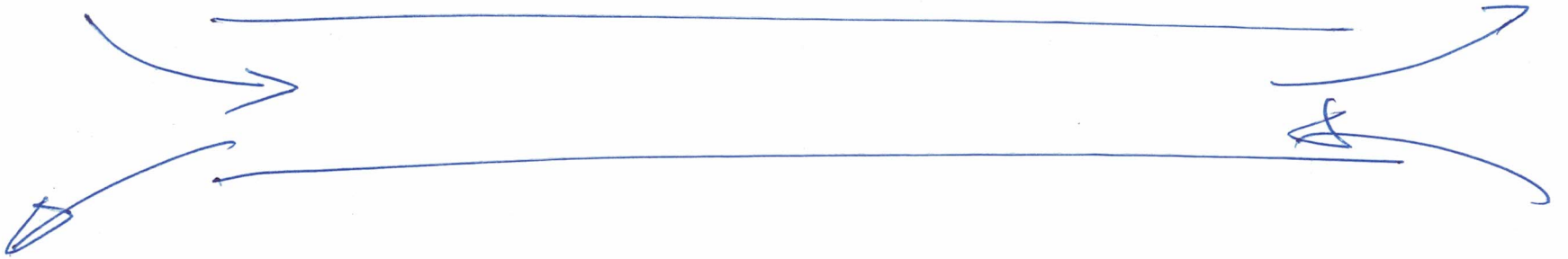
CPTS 317. 10/26

Today: Pushdown Automata (PDA)

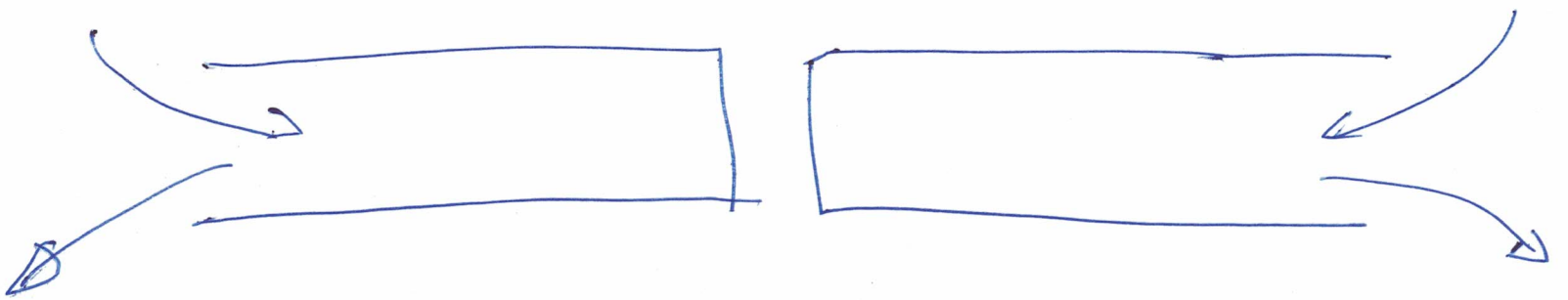
(pushdown) stack:



Queue:

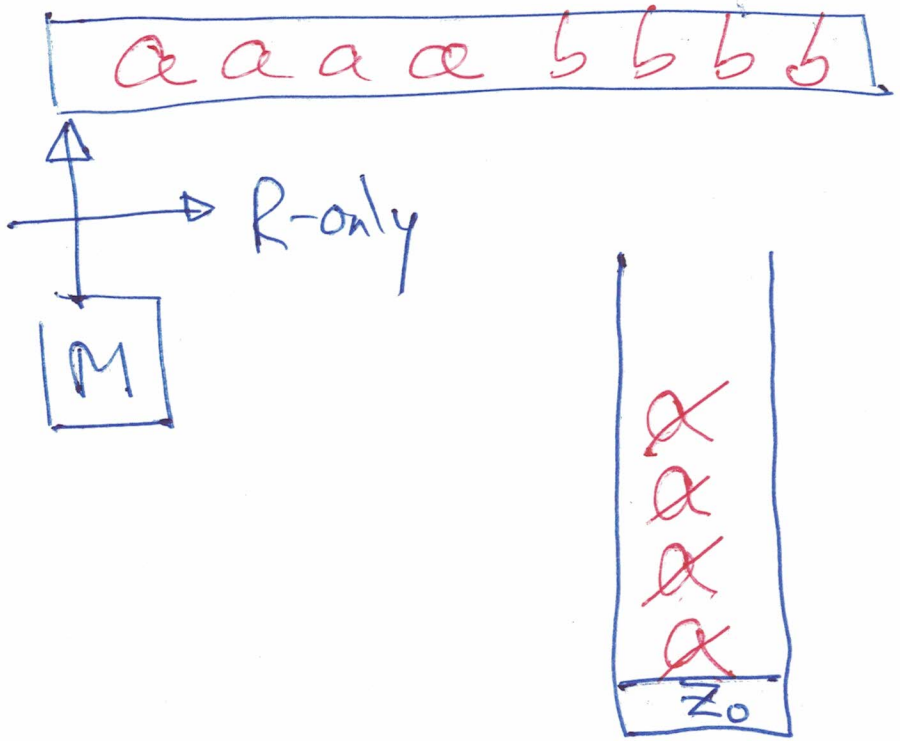


In theory, Queue = 2 stacks.



PDA = FA + pushdown steek.

Example. We know: $\{a^n b^n : n \geq 1\}$ is c.f.

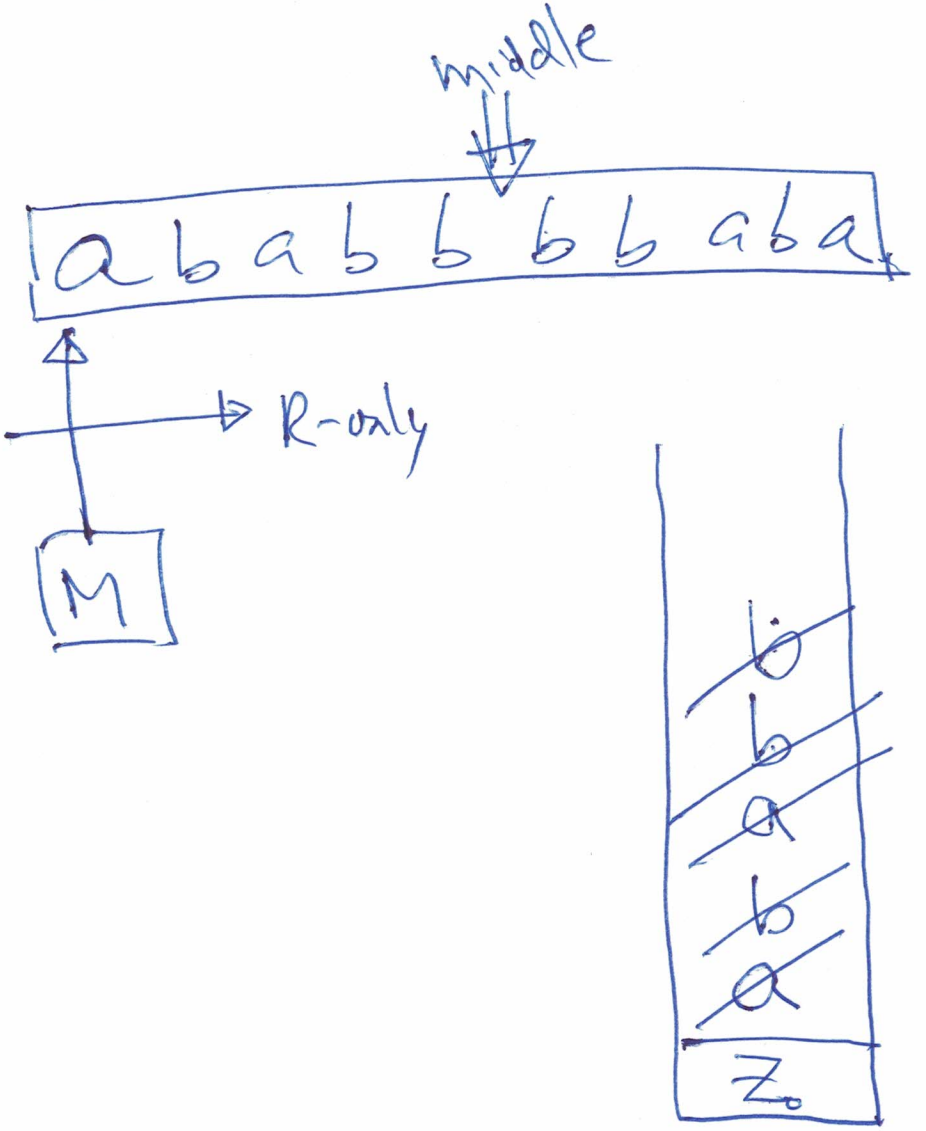


For a steek, we can push, pop, check emptyness.

I construct a PDA M to accept the language as follows. On an input word w, M keeps reading a's. For each a read, M pushes it onto the stack. Then, M keeps reading b's. For each b read, M pops an a. At the end of input w, M says yes if the stack is empty. Clearly, M accepts the language.

Note: PDA is nondet, so, we can "guess".

Exaple. Construct a PDA to accept $\{w : w = w^r, w \in \{a,b\}^*\}$.



math def of PDA.

A PDA M is a tuple

$$\langle Q, \Sigma, \Gamma, \delta, q_0, Z_0, A \rangle$$

where: Q: a finite set of states;

$q_0 \in Q$: initial state

$A \subseteq Q$: accepting/final states;

Σ: input alphabet

Γ: stack alphabet; with $Z_0 \in \Gamma$.

$$\delta : Q \times (\Sigma \cup \{\Lambda\}) \times \Gamma \longrightarrow 2^{Q \times \Gamma^*}.$$

Exaple Instructions:

$$\delta(q, a, b) = \{(p, ab)\}$$

in English: if at (current) state q, reading input symbol a, current stack top is b, then next move is: switch to state p, push a onto the stack

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$$\delta(q, a, b) = \{(p, \Lambda)\}$$

in English: if current state q, reading input symbol a, and current stack top is b, Then next move is: switch to state p, pop b (the top) out of the stack.

$$\delta(q, a, b) = \{(p, b)\}$$

do nothing on the stack.

$$\delta(q, a, b) = \underbrace{\{(p, ab), (q, \Lambda)\}}_{\text{nondet. Choose one to do next.}}.$$

M runs on input word w from its initial state and with empty stack (the stack top is Z_0)

and if there is such a run ending with an accepting state, then M accepts w.

$L(M) = \{w : M \text{ accepts } w\}$ is

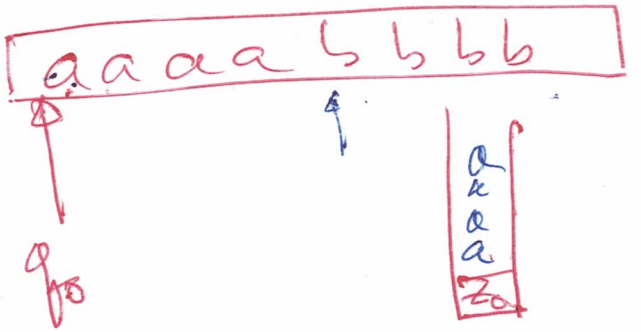
<u>the</u> language accepted by M.

Example. Construct a PDA M accepting

$$\mathcal{L} = \{a^n b^n : n \geq 1\}.$$

(English construction). 30 minutes ago.

(Explicit construction). code writing.



$\delta(q_0, a, Z_0) = \{(q_0, aZ_0)\}$ // push a to stack

$\delta(q_0, a, a) = \{(q_0, aa)\}$ //

$\delta(q_0, b, a) = \{(q_1, \Lambda)\}$ // keep popping.

$\delta(q_1, b, a) = \{(q_1, \Lambda)\}$ //

$\delta(q_1, \Lambda, Z_0) = \{(q_2, Z_0)\}$

accepting state.
(saying yes).

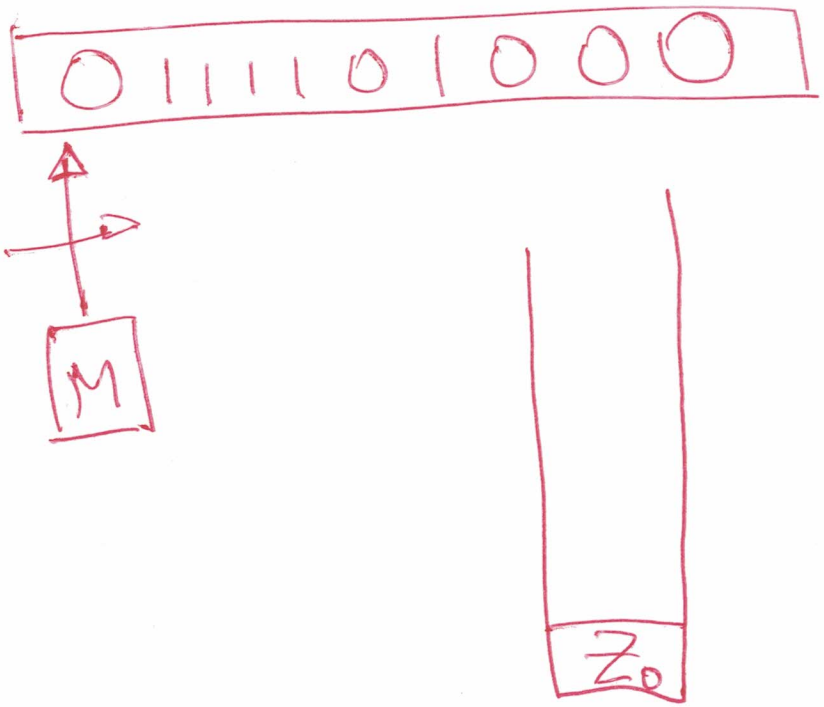
Complex Example:

Construct a PDA M that accepts

$$\mathcal{L} = \{ w \in \{0,1\}^* : \#_0(w) = \#_1(w) \}.$$

number of 0's in w.

English construction:



Thinking: I have two modes:

mode 1: see 0 → push
see 1 → pop

mode 2: see 1 → push
see 0 → pop.

difficulty: when to switch mode?
when stack is empty.